

MATLAB Examples

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By MathWorks 

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Copy and run in your MATLAB:

```
openExample('econ/DetermineCointegrationRankOfVECMoelExample')
```



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Determine Cointegration Rank of VEC Model

This example shows how to convert an n -dimensional VAR model to a VEC model, and then compute and interpret the cointegration rank of the resulting VEC model.

The rank of the error-correction coefficient matrix, C , determines the cointegration rank. If $\text{rank}(C)$ is:

- Zero, then the converted $\text{VEC}(p)$ model is a stationary $\text{VAR}(p - 1)$ model in terms of Δy_t , without any cointegration relations.
- n , then the $\text{VAR}(p)$ model is stable in terms of y_t .
- The integer r such that $0 < r < n$, then there are r cointegrating relations. That is, there are r linear combinations that comprise stationary series. You can factor the error-correction term into the two n -by- r matrices $C = \alpha\beta'$. α contains the adjustment speeds, and β the cointegration matrix. This factorization is not unique.

For more details, see [docid:econ_ug.bswxu_6-1](#) and [docid:econ_ug.brz_lcd](#), Chapter 6.3.

Consider the following $\text{VAR}(2)$ model.

$$y_t = \begin{bmatrix} 1 & 0.26 & 0 \\ -0.1 & 1 & 0.35 \\ 0.12 & -0.05 & 1.15 \end{bmatrix} y_{t-1} + \begin{bmatrix} -0.2 & -0.1 & -0.1 \\ 0.6 & -0.4 & -0.1 \\ -0.02 & -0.03 & -0.1 \end{bmatrix} y_{t-2} + \varepsilon_t.$$

Create the variables $A1$ and $A2$ for the autoregressive coefficients. Pack the matrices into a cell vector.

```
A1 = [1 0.26 0; -0.1 1 0.35; 0.12 -0.5 1.15];
A2 = [-0.2 -0.1 -0.1; 0.6 -0.4 -0.1; -0.02 -0.03 -0.1];
Var = {A1 A2};
```

Compute the autoregressive and error-correction coefficient matrices of the equivalent VEC model.

```
[Vec,C] = var2vec(Var);
```

Because the degree of the VAR model is 2, the resulting VEC model has degree $q = 2 - 1$. Hence, Vec is a one-dimensional cell array containing the autoregressive coefficient matrix.

Determine the cointegration rank by computing the rank of the error-correction coefficient matrix C .

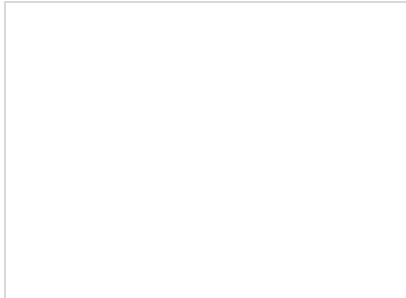
```
r = rank(C)
```

```
r =
```

```
2
```

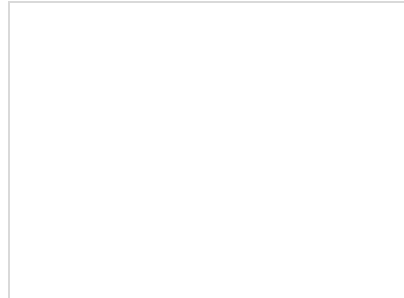
The cointegrating rank is 2. This result suggests that there are two independent linear combinations of the three variables that are stationary.

Related Examples



Test for Cointegration Using the Johansen Test

This example shows how to assess whether a multivariate time series has multiple cointegrating relations using the Johansen test. Multivariate Models



Estimate VEC Model Parameters Using egctest

This example shows how to estimate the parameters of a vector error-correction (VEC) model. Before estimating VEC model parameters, you